Linear Algebra II

FMA_2F001_EP

Lecture Notes

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These are personal lecture notes and may contain errors. For the official course material consult the lecturer.

Contents

1 Class 1: Basic Arithmetic **2**
1.1 Relations on a Set . . . 3
1.2 Modular Arithmetic and Divisibility . . . 3
1.3 Prime Numbers . . . 4

2 Lecture 2: Groups **4**

3 Lecture 3: Rings **7**
3.1 Subrings and Ideals . . . 7
3.2 Quotient Rings . . . 8
3.3 Taxonomy of Rings . . . 9
3.4 Special Ideals . . . 9

4 Lecture 4: Fields and Polynomials **10**
4.1 Polynomial Rings . . . 10
4.2 Euclidean Division and Ideals in $K[X]$ . . . 11
4.3 Roots and the Fundamental Theorem of Algebra . . . 11

5 Lecture 5: Vector Spaces **12**
5.1 Bases and Dimension . . . 13
5.2 Application: Finite Fields . . . 13
5.3 Linear Maps and Endomorphisms . . . 14

6 Lecture 6: Matrix, Reloaded **14**
6.1 Polynomials of Endomorphisms . . . 15
6.2 Eigenvalues and Eigenvectors . . . 16

7 Lecture 7: Diagonalizability **16**
7.1 Diagonalizability . . . 18

1 Class 1: Basic Arithmetic

The arithmetic fact $1 + 1 = 2$ was shown by Russel and Whitehead. The proof operates using Peano's axioms for the natural numbers $\mathbb{N}$.

Theorem 1: Peano's Axioms for $\mathbb{N}$

- **Axiom 1:** $0 \in \mathbb{N}$.
- **Axiom 2:** There exists a successor function $S : \mathbb{N} \to \mathbb{N}$.

From these, we can define $1 = S(0)$ and $2 = S(S(0))$.

Definition 2: Addition

Addition is defined by the following rules:

1. $\forall m \in \mathbb{N} : m + 0 = m$ (Add1).
2. $\forall m, n \in \mathbb{N} : m + S(n) = S(m + n)$ (Add2).

Example 3: Computing $1 + 1$

$1 + 1 = 1 + S(0) \overset{\text{(Add2)}}{=} S(1 + 0) \overset{\text{(Add1)}}{=} S(1) = 2.$

Theorem 4: Peano's Axiom 9 (Induction)

Let P be a unary predicate. To prove $\forall m \in \mathbb{N}, P(m)$ is true, it is enough to prove:

- $P(0)$ is true.
- $\forall m : P(m) \implies P(S(m))$.

Theorem 5: Well Ordering Principle (WOP)

Every non-empty subset of $\mathbb{N}$ has a minimum.

Proposition 6: WOP implies Induction

The Well Ordering Principle can be used to prove the principle of Induction.

> Proof. Let $S = \{m \mid P(m) \text{ is not true}\}$. If $P(m)$ is not true for every m, then S is non-empty. We apply WOP to S, meaning there is a minimum element $m_0 = \min S$. But $m_0 \neq 0$, so $m_0 = S(m)$ for some m. This creates a contradiction. □

1.1 Relations on a Set

Definition 7: Relations and Equivalence

Let S be a set with elements $x, y \in S$. A relation on S is defined as a subset of $S \times S$. An equivalence relation must satisfy all three of the following properties:

- **Reflexivity:** $\forall x \in S, x \sim x$.
- **Symmetry:** $\forall x, y \in S, x \sim y \iff y \sim x$.
- **Transitivity:** $\forall x, y, z \in S, x \sim y$ and $y \sim z \implies x \sim z$.

Example 8: Equivalence Relations

"Same Parity" and "Same remainder mod n" are equivalence relations.

Remark 9: Partitions

An equivalence relation is the same as a partition of S.

Definition 10: Modular Classes

For every $n \in \mathbb{N}$, $a \equiv b \pmod{n}$ defines an equivalence relation. The set of equivalence classes for this relation is denoted $\mathbb{Z}/n\mathbb{Z}$.

1.2 Modular Arithmetic and Divisibility

Theorem 11: Chinese Remainder Theorem (Sunzi)

If m and n are coprime, there is a bijection between $\mathbb{Z}/mn\mathbb{Z}$ and $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$.

Example 12: Applying the Chinese Remainder Theorem

For $n = 2$ and $m = 3$, we can map between $\mathbb{Z}/6\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. If we want to find all $m \in \mathbb{Z}$ such that $m \equiv 2 \pmod{5}$ and $m \equiv 3 \pmod{7}$, the solution is the set $\{14 + 35k \mid k \in \mathbb{Z}\}$.

Theorem 13: Euclidean Division

$\forall a, \forall b \neq 0, \exists (q, r)$ such that $a = bq + r$ and $0 \leq r < |b|$.

Theorem 14: Bézout's Theorem

$\exists (u, v) \in \mathbb{Z}^2$ such that $au + bv = \gcd(a, b)$.

Example 15: Euclid's Algorithm

Applying the algorithm to find coefficients for 5 and 7:

- $7 = 1 \times 5 + 2$.
- $5 = 2 \times 2 + \gcd(5, 7)$.
- $1 = 5 - 2 \times 2 = 5 - 2 \times (7 - 5) = 3 \times 5 - 2 \times 7$.

We have found that $3 \times 5 \to (0, 1)$ and $-2 \times 7 \to (1, 0)$. If we want to find m such that $m \equiv 2 \pmod 5$ and $m \equiv 3 \pmod 7$, we compute $(3 \times 5) \times 3 + (-2 \times 7) \times 2$.

1.3 Prime Numbers

Definition 16: Prime Number

A prime number is a number $p \in \mathbb{N}$ such that $p \geq 2$ and p has only divisors 1 and p.

Theorem 17: Infinitude of Primes

There are infinitely many primes.

Theorem 18: Tao's Theorem

At least one of the following sets of primes (p, q) is infinite:

- **Twin primes:** $|p - q| = 2$.
- **Cousin primes:** $|p - q| = 4$.
- **Sexy primes:** $|p - q| = 6$.

Remark 19: Goldbach Conjecture

The Goldbach conjecture posits that every even number is equal to $p + q$ with p and q primes.

2 Lecture 2: Groups

Definition 1: Group

A group is a set G equipped with a binary operation (or group law) $*$, denoted $(G, *)$, that satisfies the following properties:

1. **Associativity:** $\forall a, b, c \in G, (a * b) * c = a * (b * c)$.
2. **Neutral Element:** $\exists e \in G$ such that $\forall a \in G, a * e = e * a = a$.
3. **Inverse Element:** $\forall a \in G, \exists b \in G$ such that $a * b = b * a = e$.

Optional: If a group also satisfies commutativity (i.e., $\forall a, b \in G, a * b = b * a$), it is called an **Abelian group**.

Example 2: Groups in Modular Arithmetic

- The set $\mathbb{Z}/n\mathbb{Z}$ with addition is a group. The neutral element is $[0]_n$ and the inverse of $[k]_n$ is $[-k]_n$.
- If p is prime, the non-zero elements modulo p under multiplication form a group. To find the inverse of k modulo p, one can use Bézout's identity on k and p. Because p is prime, there exist integers u and v such that $uk + vp = 1$. Modulo p, this simplifies to $uk \equiv 1 \pmod{p}$, which implies that u is the multiplicative inverse k^{-1}.

Definition 3: Powers and Cyclic Groups

For an element $g \in G$, its powers are defined as follows:

- For $n > 0$, $g^n = g * g * \cdots * g$ (n times).
- For $n = 0$, $g^0 = e$.
- For $n < 0$, $g^n = g^{-1} * g^{-1} * \cdots * g^{-1}$ ($|n|$ times).

A finite group G is called **cyclic** if it is generated by a single element g, meaning $G = \langle g \rangle = \{g^m \mid m \in \mathbb{Z}\}$.

Example 4: Common Groups

- **Dihedral groups** D_k: the group of symmetries of a regular k-gon.
- **Symmetric groups** Σ_n (or S_n): the group of permutations of n elements.
- **Alternating group** A_n: the subgroup of Σ_n containing all even permutations, i.e., $A_n = \{\sigma \in \Sigma_n \mid \varepsilon(\sigma) = 1\}$.
- The group of symmetries of polyhedra, such as the Rhombic Triacontahedron.

Definition 5: Subgroups and Morphisms

- A **subgroup** is a subset of G that forms a group under the same group law.
- A **morphism** (or homomorphism) is a map φ between groups that preserves the group structure, e.g., $\varphi(a * b) = \varphi(a) * \varphi(b)$.
- An **isomorphism** is a bijective group morphism.

Definition 6: Order of an Element

The order of an element $g \in G$ is the smallest positive integer m such that $g^m = e$. It is denoted $\omega(g) = \min\{m \geq 1 \mid g^m = e\}$.

Theorem 7: Weak Lagrange Theorem

Let G be a finite Abelian group, and let $g \in G$. Then the order of g divides the size of the group, i.e., $\omega(g) \mid \#G$.

PROOF. Let $\pi = \prod_{h \in G} h$ be the product of all elements in G. Consider the map $\varphi_g : G \to G$ defined by $h \mapsto g * h$. This map is a bijection, as it has an inverse $\varphi_{g^{-1}}$. Because G is Abelian, we can rearrange the product:

$$\pi = \prod_{h \in G} h = \prod_{h \in G} \varphi_g(h) = \prod_{h \in G} (g * h)$$

Factoring out g for each of the $\#G$ elements, we get $\pi = g^{\#G} \cdot \prod_{h \in G} h = g^{\#G} \cdot \pi$. By canceling π, we find $g^{\#G} = e$.
We can write $\#G = q \cdot \omega(g) + r$ using Euclidean division, with $0 \leq r < \omega(g)$. Substituting this into our equation gives:

$$e = g^{\#G} = g^{q\omega(g)+r} = (g^{\omega(g)})^q \cdot g^r = e^q \cdot g^r = g^r$$

Since $g^r = e$ and $0 \leq r < \omega(g)$, the minimality of the order implies $r = 0$. Therefore, $\omega(g)$ divides $\#G$. □

Theorem 8: Strong Lagrange Theorem

If G is a finite group and $H \subset G$ is a subgroup, then $\#H$ divides $\#G$.

PROOF. We say two elements $x, y \in G$ are equivalent modulo H ($x \sim y$) if $y^{-1}x \in H$. This condition is equivalent to $x \in yH$, where yH represents a left coset. The left cosets form a partition of the group G, so $G = \bigsqcup_{y_i \in I} y_i H$ for some index set I of representatives.
Every coset $y_i H$ has exactly the same size, which is equal to $\#H$. Thus, we can conclude that the total number of elements in G is given by $\#G = \#H \times \#I$. This immediately shows that $\#H \mid \#G$. □

Theorem 9: Cauchy's Theorem

Let G be a finite group and let p be a prime number such that $p \mid \#G$. Then there exists an element $g \in G$ such that $\omega(g) = p$.

PROOF. Consider the set $S_p = \{(g_1, g_2, \ldots, g_p) \mid g_i \in G, g_1 g_2 \ldots g_p = e\}$. We define an equivalence relation on S_p using cyclic shifts: two tuples are equivalent if one is a cyclic permutation of the other (i.e., shifting the sequence by some offset k). Notice that the size of S_p is $(\#G)^{p-1}$, because the first $p-1$ elements can be chosen freely, and the final element g_p is uniquely determined as $(g_1 \ldots g_{p-1})^{-1}$.
Under this equivalence relation, every equivalence class contains either 1 or p elements. The tuple $(e, \ldots, e)$ forms an equivalence class of size 1. We can partition S_p into these classes, yielding the equation $\#S_p = N_1 + pN_p$, where N_1 is the number of classes of size 1, and N_p is the number of classes of size p.
Since $p \mid \#G$, we know $p \mid (\#G)^{p-1}$, so $p \mid \#S_p$. Thus, $p \mid (N_1 + pN_p)$, which implies $p \mid N_1$. We already know $N_1 \geq 1$ because of the class containing $(e, \ldots, e)$. Because $p \mid N_1$, we must have $N_1 \geq p \geq 2$.
This means there is at least one other class of size 1, which corresponds to a tuple $(g, g, \ldots, g) \in S_p$ for some $g \neq e$. Since this tuple is in S_p, it must satisfy $g^p = e$. Because p is prime and $g \neq e$, the order of g is exactly p, so $\omega(g) = p$. □

3 Lecture 3: Rings

Definition 1: Unital Ring

A unital ring is a set R equipped with two binary operations, denoted $(R, +, \times)$, satisfying the following properties:

1. $(R, +)$ is an Abelian group, with a neutral element denoted 0 or 0_R.
2. Multiplication "$\times$" is associative: $\forall x, y, z \in R, x(yz) = (xy)z$.
3. Multiplication "$\times$" has a neutral element, denoted 1 or 1_R.
4. Distributivity of multiplication over addition: $\forall a, b, c \in R, a(b + c) = ab + ac$ and $(b + c)a = ba + ca$.

Note: $(R, \times)$ and $(R\backslash\{0\}, \times)$ are not necessarily groups, because elements do not generally have multiplicative inverses.
If the multiplication operation is also commutative, then R is called a **commutative ring**.

Definition 2: Ring Morphism

A map $\varphi : (R_1, +, \times) \to (R_2, +, \times)$ is a ring morphism if:

1. $\varphi : (R_1, +) \to (R_2, +)$ is a group morphism.
2. $\varphi(1_{R_1}) = 1_{R_2}$.
3. $\forall a, b \in R_1, \varphi(ab) = \varphi(a)\varphi(b)$.

Theorem 3: The Ring of Integers as an Initial Object

The ring $(\mathbb{Z}, +, \times)$ is the paradigmatic example of a ring, and it serves as the initial object in the category of unital rings. This means that for any ring R, there exists a unique ring morphism $\psi : \mathbb{Z} \to R$. Therefore, $\mathbb{Z}$ is the unique initial object up to ring isomorphism.

Proof. By "abstract nonsense", let R_0 be another initial object. Then there exist unique morphisms $\psi_1 : \mathbb{Z} \to R_0$ and $\psi_2 : R_0 \to \mathbb{Z}$. The composition $\psi_2 \circ \psi_1 : \mathbb{Z} \to \mathbb{Z}$ must be the identity morphism (since $\mathbb{Z}$ is initial), which implies $\psi_1 = \psi_2^{-1}$, establishing a ring isomorphism between $\mathbb{Z}$ and R_0. □

3.1 Subrings and Ideals

Definition 4: Subring

A subring is a subset that forms a ring under the same (induced) algebraic structure. For example, we have the chain of subrings: $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C} \subset \mathbb{H}$.

Remark 5: Ideals vs. Subrings

For groups, the kernel of a group morphism is a subgroup. However, for rings, the kernel is generally *not* a subring (it usually does not contain 1_R). The correct notion to capture

the behavior of kernels in rings is an **ideal**.

Definition 6: Ideal

An ideal $I \subset R$ is a subgroup of $(R, +)$ that is stable under multiplication by any element of R. Equivalently: $\forall a, b \in R$ and $\forall x, y \in I$, we have $ax + by \in I$ (analogous to a linear combination in a vector space).

Proposition 7: Kernels are Ideals

Let $\varphi : R_1 \to R_2$ be a ring morphism. Then $\ker(\varphi) = \{x \in R_1 \mid \varphi(x) = 0\}$ is an ideal of R_1 (but not a subring!).

PROOF. Let $a, b \in R_1$ and $x, y \in \ker(\varphi)$. We want to prove that $ax + by \in \ker(\varphi)$, which means $\varphi(ax + by) = 0$. Using the properties of ring morphisms:

$$\varphi(ax + by) = \varphi(ax) + \varphi(by) = \varphi(a)\varphi(x) + \varphi(b)\varphi(y)$$

Since $x, y \in \ker(\varphi)$, we know $\varphi(x) = 0$ and $\varphi(y) = 0$. Thus, $\varphi(a) \cdot 0 + \varphi(b) \cdot 0 = 0$. □

Definition 8: Principal Ideal

In a commutative ring R, ideals of the form $(x) = xR = \{xy \mid y \in R\}$ are called **principal ideals**. For example, all ideals of $\mathbb{Z}$ are principal (of the form $m\mathbb{Z}$).

3.2 Quotient Rings

Theorem 9: Quotient Ring Construction

For any commutative ring R and any ideal $I \subset R$, we can construct a quotient space R/I by defining an equivalence relation $r_1 \sim r_2 \iff r_1 - r_2 \in I$. The equivalence classes (or cosets) are denoted $[r] = r + I$.
The set of these classes, R/I, inherits a natural ring structure given by:

- $[r_1] + [r_2] = [r_1 + r_2]$
- $[r_1] \times [r_2] = [r_1 \times r_2]$

Example 10: Modular Arithmetic and CRT

Let $R = \mathbb{Z}$ and $I = m\mathbb{Z}$. This construction gives the usual quotient ring $\mathbb{Z}/m\mathbb{Z}$. If m and n are coprime, the Chinese Remainder Theorem establishes a ring isomorphism: $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$.

3.3 Taxonomy of Rings

Definition 11: Types of Rings

- **Integral Domain:** A commutative ring where $xy = 0 \implies x = 0$ or $y = 0$ (i.e., it has no zero divisors).
- **Principal Ideal Domain (PID):** An integral domain where every ideal is a principal ideal.
- **Division Ring:** A ring where $(R \setminus \{0\}, \times)$ is a group (i.e., every non-zero element has a multiplicative inverse).
- **Field:** A commutative division ring (i.e., $(R \setminus \{0\}, \times)$ is an Abelian group).

Remark 12: Examples and Hierarchy

We have the following strict inclusions for commutative rings:

$$\text{Commutative Rings} \supset \text{Integral Domains} \supset \text{PIDs} \supset \text{Fields}$$

- **Non-commutative Rings:** $M_2(\mathbb{R}), M_3(\mathbb{R})$.
- **Division Rings (not fields):** The quaternions $\mathbb{H}$.
- **Fields:** $\mathbb{R}, \mathbb{Q}, \mathbb{Z}/p\mathbb{Z}$ (where p is prime).

3.4 Special Ideals

Definition 13: Prime and Maximal Ideals

Let $I \subsetneq R$ be a proper ideal in a commutative ring R.

- I is a **prime ideal** if $xy \in I \implies x \in I$ or $y \in I$. For example, $2\mathbb{Z}$ is a prime ideal, but $4\mathbb{Z}$ is not.
- I is a **maximal ideal** if for any ideal J such that $I \subseteq J \subsetneq R$, it must be that $I = J$.

Theorem 14: Quotients by Prime and Maximal Ideals

Let R be a commutative ring and I be a proper ideal.

- I is a maximal ideal $\iff$ $(R/I, +, \times)$ is a field.
- I is a prime ideal $\iff$ $(R/I, +, \times)$ is an integral domain.

Because every field is an integral domain, this immediately implies that every maximal ideal is also a prime ideal.

4 Lecture 4: Fields and Polynomials

Definition 1: Field

A set K equipped with two operations $(K, +, \times)$ is a field if:

1. It is a commutative ring.
2. It is a division ring (i.e., every non-zero element has a multiplicative inverse, so $\forall x \neq 0, \exists y$ such that $xy = 1$).

Remark 2: Properties of Fields

- The non-zero elements form a group under multiplication, denoted $(K^*, \times)$.
- Every field $(K, +, \times)$ is an integral domain (since $xy = 0 \iff x = 0$ or $y = 0$).

Definition 3: Field Extensions

A **subfield** is a subset of a field that is itself a field under the induced operations. If $K \subset L$ is a subfield, we usually refer to L as a **field extension** of K. Field extensions often arise to supply "missing" roots to equations. For example, $\mathbb{C}$ extends $\mathbb{R}$ to provide roots for $X^2 + 1 = 0$, and $\mathbb{R}$ extends $\mathbb{Q}$ to provide roots for polynomials like $X^2 - 2$, $X^2 - 3$, or $X^3 - 4$.

4.1 Polynomial Rings

Definition 4: The Polynomial Ring $K[X]$

Let K be a field. The set of polynomials with coefficients in K is denoted $K[X]$. An element $P \in K[X]$ is of the form $P(X) = a_0 + a_1 X + \cdots + a_n X^n$, where $(a_0, \ldots, a_n)$ is a finite sequence of elements in K.

- $(K[X], +, \times)$ is a commutative ring, but **not** a division ring.
- If $a_n \neq 0$, then the **degree** of P is $\deg(P) = n$.
- By convention, the zero polynomial $P = 0$ has degree $\deg(0) = -\infty$.

Proposition 5: Degree and Integral Domain

For any polynomials $P, Q \in K[X]$:

1. $\deg(PQ) = \deg(P) + \deg(Q)$.
2. $K[X]$ is an integral domain.

Proof. If $PQ = 0$, then $\deg(PQ) = -\infty$. Therefore, $\deg(P) + \deg(Q) = -\infty$, which requires either $\deg(P) = -\infty$ or $\deg(Q) = -\infty$. Thus, $P = 0$ or $Q = 0$. □

Remark 6: Polynomials as Functions

Every polynomial $P \in K[X]$ can be identified with a polynomial function $K \to K$, mapping $\alpha \mapsto P(\alpha) = a_0 + a_1\alpha + \cdots + a_n\alpha^n$. An element $\alpha \in K$ is called a **root** of P if $P(\alpha) = 0$.

4.2 Euclidean Division and Ideals in $K[X]$

Theorem 7: Euclidean Division in $K[X]$

For any polynomials $A, B \in K[X]$ with $B \neq 0$, there exist unique polynomials $Q, R \in K[X]$ such that:

$$A(X) = B(X)Q(X) + R(X) \quad \text{with} \quad \deg(R) < \deg(B)$$

Example 8: Computing Euclidean Division

Let $A(X) = X^2 + 3X + 1$ and $B(X) = 2X + 2$ in $\mathbb{Q}[X]$. We can rewrite A as:

$$X^2 + 3X + 1 = (2X + 2)\left(\frac{1}{2}X + 1\right) - 1$$

Here, the quotient is $Q(X) = \frac{1}{2}X + 1$ and the remainder is $R(X) = -1$.

Theorem 9: $K[X]$ is a Principal Ideal Domain (PID)

Every ideal in $K[X]$ is principal.

Proof. Let $I \subset K[X]$ be a non-zero ideal. We want to show that $I = (B) = B \cdot K[X]$ for some $B \in I$. If I contains a non-zero constant polynomial, then $1 \in I \implies I = K[X]$, which is generated by the constant polynomial 1.
Otherwise, by the Well Ordering Principle, let $B \in I$ be a non-zero polynomial of smallest degree in I. Let $A \in I$ be any polynomial. We perform Euclidean division of A by B:

$$A = BQ + R \quad \text{where} \quad \deg(R) < \deg(B)$$

Since $A \in I$ and $B \in I$, and I is an ideal, the remainder $R = A - BQ$ must also be in I. However, $\deg(R) < \deg(B)$, and B was chosen to have the minimal degree among non-zero elements in I. This implies R must be the zero polynomial ($R = 0$). Therefore, $A = BQ$, meaning every element in I is a multiple of B. Thus, $I = (B)$. □

4.3 Roots and the Fundamental Theorem of Algebra

Proposition 10: Roots correspond to Linear Factors

If $\alpha \in K$ is a root of $P \in K[X]$, then $P(X) = (X - \alpha)Q(X)$ for some $Q \in K[X]$.

Proof. Perform Euclidean division of $P(X)$ by $(X - \alpha)$:

$$P(X) = (X - \alpha)Q(X) + R(X)$$

The remainder must satisfy $\deg(R) < \deg(X - \alpha) = 1$, which means $\deg(R) \leq 0$. Thus, $R(X)$ is a constant polynomial r. Evaluating the polynomial at α gives $P(\alpha) = (\alpha - \alpha)Q(\alpha) + r = r$.

Since α is a root, $P(\alpha) = 0$, so $r = 0$. Therefore, $P(X) = (X - \alpha)Q(X)$. □

Corollary 11: Maximum Number of Roots

A non-zero polynomial $P \in K[X]$ of degree n has at most n roots in K.

Proof. For every root found, the polynomial can be factored by a degree 1 factor, reducing the degree of the quotient by 1. This process can be repeated at most n times. □

Theorem 12: Fundamental Theorem of Algebra

(Referred to in lecture as an "insignificant analytic lemma")
Every polynomial $P \in \mathbb{C}[X]$ with $\deg(P) \geq 1$ has at least one root in $\mathbb{C}$.

Corollary 13: Factorization in $\mathbb{C}[X]$

Every polynomial $P \in \mathbb{C}[X]$ of degree $n \geq 1$ has exactly n roots in $\mathbb{C}$ (counted with multiplicity). It factors completely into linear terms:

$$P(X) = c(X - z_1)(X - z_2)\dots(X - z_n)$$

where c is the leading coefficient.

Proof Sketch of the Fundamental Theorem of Algebra. Let $P(z) = a_0 + a_1 z + \cdots + a_n z^n$ with $a_n \neq 0$ and $n \geq 1$. If $a_0 = 0$, then $P(0) = 0$, so $z = 0$ is a root. Assume $a_0 \neq 0$.
Consider the image of a circle of radius r under the polynomial map, parameterized by $z = re^{i\theta}$. For a very large radius r, the highest degree term dominates, meaning $P(re^{i\theta}) \approx a_n r^n e^{in\theta}$. This creates a large curve that winds around the origin n times. As $r \to 0$, the curve shrinks continuously to the point $P(0) = a_0$, which is non-zero (so it no longer winds around the origin). By continuity, as the curve deforms from winding around the origin to a point away from the origin, the curve must cross 0 at some radius r and angle θ. At this intersection, $P(re^{i\theta}) = 0$, giving a root. □

5 Lecture 5: Vector Spaces

Let K be a field (usually $K = \mathbb{R}, \mathbb{C},$ or $\mathbb{Q}$). Recall that $(K, +)$ is an Abelian group and $(K\backslash\{0\}, \times)$ is an Abelian group. The elements of K are called **scalars**.

Definition 1: Vector Space

A set E equipped with an internal addition "+" and an external scalar multiplication "·" is a K-vector space, denoted $(E, +, \cdot)$, if:

1. $(E, +)$ is an Abelian group.
2. $\forall \lambda \in K, \forall x, y \in E, \lambda \cdot (x + y) = \lambda \cdot x + \lambda \cdot y.$
3. $\forall \lambda, \mu \in K, \forall x \in E, (\lambda + \mu) \cdot x = \lambda \cdot x + \mu \cdot x.$
4. $\forall \lambda, \mu \in K, \forall x \in E, \lambda \cdot (\mu \cdot x) = (\lambda\mu) \cdot x.$
5. $\forall x \in E, 1_K \cdot x = x.$

5.1 Bases and Dimension

Definition 2: Span, Independence, and Basis

Let $\{e_1, \dots, e_k\}$ be a family of vectors in E.

- The **span** of this family is the set of all their finite linear combinations: $\langle e_1, \dots, e_k \rangle = \operatorname{span}(e_1, \dots, e_k)$.
- The family **generates** (or spans) E if $\operatorname{span}(e_1, \dots, e_k) = E$.
- The family is **linearly independent** (or free) if $\sum_{i=1}^{k} \lambda_i e_i = 0 \implies \lambda_i = 0$ for all i. Otherwise, they are linearly dependent (or linked).
- A **basis** of E is a family of vectors that is both free and generates E.

Remark 3: Facts from Linear Algebra I

- If E is generated by finitely many vectors, then it has a basis containing finitely many vectors.
- All bases of E have the same cardinality, which is defined as the dimension of E, denoted $N = \dim_K E$.
- In a finite-dimensional setting ($N = \dim_K E < \infty$), every linearly independent family of vectors can be extended into a basis (Incomplete Basis Theorem).

Example 4: Dimensions over different fields

- $\dim_{\mathbb{R}} \mathbb{C} = 2$ (Basis: $\{1, i\}$).
- $\dim_{\mathbb{C}} \mathbb{C} = 1$ (Basis: $\{1\}$).
- $\dim_{\mathbb{Q}} \mathbb{C} = \infty$.

5.2 Application: Finite Fields

We know that $\mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p$ is a finite field when p is prime.

Theorem 5: Cardinality of Finite Fields

Any finite field L has a cardinality of p^d, where p is a prime number and $d \geq 1$ is an integer.

Proof. Consider the unique ring morphism $\varphi : \mathbb{Z} \to L$. Since L is finite, the kernel of this morphism is non-trivial, meaning there exists a smallest non-zero integer m such that $\varphi(m) = 0$ in L (e.g., $1 + 1 + \cdots + 1 = 0$). Because L is an integral domain, this integer m must be a prime number p, which is called the **characteristic** of the field. This implies that $\mathbb{Z}/p\mathbb{Z}$ is isomorphic to a subfield of L, so $\mathbb{Z}/p\mathbb{Z} \subset L$.
Now, consider L as a vector space over the field $\mathbb{Z}/p\mathbb{Z}$. Since L is finite, it must have a finite dimension d. Let $\{e_1, \dots, e_d\}$ be a basis for L over $\mathbb{Z}/p\mathbb{Z}$. Every element $x \in L$ can be written uniquely as $x = \lambda_1 e_1 + \cdots + \lambda_d e_d$, where each $\lambda_i \in \mathbb{Z}/p\mathbb{Z}$. Since there are d coefficients and p choices for each coefficient, the total number of elements in L is $|L| = p^d$. □

Remark 6: Uniqueness of Finite Fields

Up to isomorphism, there is exactly one field of cardinality p^d, typically denoted as $\mathbb{F}_{p^d}$.

5.3 Linear Maps and Endomorphisms

Let E_1 and E_2 be two vector spaces over the same field K.

Definition 7: Linear Maps

- A function $f : E_1 \to E_2$ is a **linear map** if it preserves the vector space structure: $\forall x, y \in E_1, \forall \lambda \in K, f(\lambda x + y) = \lambda f(x) + f(y)$.
- An **endomorphism** is a linear map from a space to itself ($f : E_1 \to E_1$).
- An **isomorphism** is a bijective linear map.
- An **automorphism** is an endomorphism that is also an isomorphism (bijective).

The space of all endomorphisms of E is denoted $L(E)$ or $\mathcal{L}(E)$.

Proposition 8: Structure of $L(E)$

- $(L(E), +, \cdot)$ is a K-vector space. If $\dim_K E = N$, then $\dim_K L(E) = N^2$.
- The composition of maps "$\circ$" acts as a non-commutative multiplication law on $L(E)$. For $f, g \in L(E)$, $f \circ g \in L(E)$.
- $(L(E), +, \circ)$ is a ring.
- Combined, $(L(E), +, \circ, \cdot)$ forms what is called an **algebra**.

Theorem 9: Rank-Nullity Theorem

Let $f : E_1 \to E_2$ be a linear map where E_1 and E_2 are finite-dimensional. Let $\mathrm{Rk}(f) = \dim(\mathrm{Im}(f))$ and $\mathrm{Null}(f) = \dim(\ker(f))$. Then:

$$\mathrm{Rk}(f) + \mathrm{Null}(f) = \dim(E_1)$$

6 Lecture 6: Matrix, Reloaded

Let $\mathbb{K}$ be a field (e.g., $\mathbb{R}, \mathbb{C}, \mathbb{Q}$) and let E be a vector space over $\mathbb{K}$ of finite dimension N. Consider an endomorphism $f \in \mathcal{L}(E)$ (a linear map from E to E).

Definition 1: Matrix of an Endomorphism

Let $\mathcal{E} = \{e_1, \ldots, e_N\}$ be a basis of E. We can represent any vector $x = \sum_{i=1}^{N} \alpha_i e_i \in E$ as a column vector of its coordinates $(\alpha_1, \ldots, \alpha_N)^T$.
The matrix of $f \in \mathcal{L}(E)$ in the basis $\mathcal{E}$ is formed by evaluating f on each basis vector e_j, and writing the result as a column vector:

$$M = \mathrm{Mat}_{\mathcal{E}}(f) = [f(e_1) \mid f(e_2) \mid \cdots \mid f(e_N)]$$

If $f(e_j) = a_{1j}e_1 + a_{2j}e_2 + \cdots + a_{Nj}e_N$, then the resulting matrix is:

$$M = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1N} \\ a_{21} & a_{22} & \dots & a_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ a_{N1} & a_{N2} & \dots & a_{NN} \end{pmatrix}$$

Here, a_{ij} is the coefficient (or entry) on row i and column j.

Remark 2: Matrix Algebra and Multiplication

The set of $N \times N$ matrices with coefficients in $\mathbb{K}$, denoted $\mathcal{M}_N(\mathbb{K})$, forms an algebra $(\mathcal{M}_N(\mathbb{K}), +, \cdot, \times)$. Matrix multiplication $C = AB$ is defined by the formula $C_{ij} = \sum_{k=1}^{N} A_{ik}B_{kj}$. This directly corresponds to the composition of endomorphisms: $\mathrm{Mat}_{\mathcal{E}}(f \circ g) = \mathrm{Mat}_{\mathcal{E}}(f) \times \mathrm{Mat}_{\mathcal{E}}(g)$.
When computing AB, remember that multiplication on the left acts on the columns of the right matrix:

$$A[B_1 \mid B_2 \mid \cdots \mid B_N] = [AB_1 \mid AB_2 \mid \cdots \mid AB_N]$$

6.1 Polynomials of Endomorphisms

Definition 3: Polynomial of an Endomorphism

For an endomorphism $u \in \mathcal{L}(E)$, its powers are defined by repeated composition: $u^m = u \circ u \circ \cdots \circ u$ (m times). By convention, $u^0 = \mathrm{Id}$.
If we consider a polynomial $P \in \mathbb{K}[X]$, given by $P(X) = \sum_{k=0}^{m} a_k X^k$, we can "evaluate" P at u by substituting X with u:

$$P(u) = \sum_{k=0}^{m} a_k u^k$$

Theorem 4: The Minimal Polynomial

The evaluation map $\mathrm{ev}_u : \mathbb{K}[X] \to \mathcal{L}(E)$ given by $P \mapsto P(u)$ is a ring morphism. Because the dimension of $\mathbb{K}[X]$ is infinite and the dimension of $\mathcal{L}(E)$ is finite (N^2), this morphism cannot be injective. Therefore, it has a non-trivial kernel.
The kernel, $\ker(\mathrm{ev}_u)$, is an ideal of $\mathbb{K}[X]$. Since $\mathbb{K}[X]$ is a Principal Ideal Domain (PID), this ideal is generated by a unique monic polynomial. This unique monic polynomial is called the **minimal polynomial** of u, denoted P_u.
Equivalently:

- P_u is the monic polynomial of smallest degree that cancels u (i.e., $P_u(u) = 0$).
- Every polynomial P that cancels u is a multiple of P_u.

6.2 Eigenvalues and Eigenvectors

Definition 5: Spectrum and Eigenspaces

Let $\lambda \in \mathbb{K}$, $x \in E$ with $x \neq 0$, and $u \in \mathcal{L}(E)$. If $u(x) = \lambda x$, then λ is called an **eigenvalue** of u, and x is called an **eigenvector** associated with λ.
The **eigenspace** associated with λ is defined as:

$$E_\lambda = \{x \in E \mid u(x) = \lambda x\} = \ker(u - \lambda \mathrm{Id})$$

The set of all eigenvalues of u is called the **spectrum** of u, denoted $\mathrm{Sp}(u)$.

Example 6: Spectra and Minimal Polynomials

Let $u = \alpha \mathrm{Id}$ in $\mathcal{L}(\mathbb{R}^3)$. In any basis, its matrix is $\mathrm{diag}(\alpha, \alpha, \alpha)$. The minimal polynomial is $P_u(X) = X - \alpha$ (since $u - \alpha \mathrm{Id} = 0$). The spectrum is $\mathrm{Sp}(u) = \{\alpha\}$, which matches the roots of P_u.

Proposition 7: Polynomial of an Eigenvalue

If $\lambda \in \mathrm{Sp}(u)$, then for any polynomial $P \in \mathbb{K}[X]$, $P(\lambda) \in \mathrm{Sp}(P(u))$.

Proof. By definition, there exists a non-zero vector $x \in E$ such that $u(x) = \lambda x$. By induction, $u^k(x) = \lambda^k x$ for any integer $k \geq 0$. Let $P(X) = \sum_{k=0}^{m} a_k X^k$. We apply $P(u)$ to x:

$$P(u)(x) = \left(\sum_{k=0}^{m} a_k u^k\right)(x) = \sum_{k=0}^{m} a_k u^k(x) = \sum_{k=0}^{m} a_k \lambda^k x = \left(\sum_{k=0}^{m} a_k \lambda^k\right) x = P(\lambda)x$$

Since $P(u)(x) = P(\lambda)x$ and $x \neq 0$, $P(\lambda)$ is an eigenvalue of $P(u)$. □

Proposition 8: Roots of the Minimal Polynomial

If $\lambda \in \mathrm{Sp}(u)$, then $P_u(\lambda) = 0$. In other words, the eigenvalues of u are roots of its minimal polynomial.

Proof. Assume $\lambda \in \mathrm{Sp}(u)$. By the previous proposition applied to the minimal polynomial P_u, we must have $P_u(\lambda) \in \mathrm{Sp}(P_u(u))$. However, by the definition of the minimal polynomial, $P_u(u) = 0$ (the zero endomorphism). The spectrum of the zero endomorphism is simply $\{0\}$. Therefore, $P_u(\lambda) = 0$. □

7 Lecture 7: Diagonalizability

Let K be a field and E be a vector space over K of finite dimension N. Let $\mathcal{E} = \{e_1, \ldots, e_N\}$ be a basis of E, and let $u \in \mathcal{L}(E)$ be an endomorphism with matrix $M = \mathrm{Mat}_{\mathcal{E}}(u)$.

Theorem 1: Spectrum and Minimal Polynomial

The spectrum of u is exactly the set of roots of its minimal polynomial:

$$\mathrm{Sp}(u) = \{\lambda \in K \mid P_u(\lambda) = 0\}$$

PROOF. We have already proven the forward inclusion ($\lambda \in \mathrm{Sp}(u) \implies P_u(\lambda) = 0$) in the previous lecture. To prove the reverse, let λ be a root of P_u. This means we can factor the minimal polynomial as $P_u(X) = (X - \lambda)Q(X)$.
Evaluating at the endomorphism u, we get the zero endomorphism:

$$0_{\mathcal{L}(E)} = P_u(u) = (u - \lambda\mathrm{Id}) \circ Q(u)$$

Because P_u is the polynomial of *minimal* degree that cancels u, and $\deg(Q) < \deg(P_u)$, it must be that $Q(u) \neq 0_{\mathcal{L}(E)}$. Thus, there exists some vector $x \in E$ such that $y = Q(u)(x) \neq 0_E$.
Applying $(u - \lambda\mathrm{Id})$ to y, we get:

$$(u - \lambda\mathrm{Id})(y) = (u - \lambda\mathrm{Id})(Q(u)(x)) = P_u(u)(x) = 0_E$$

This shows that $y \in \ker(u - \lambda\mathrm{Id})$ with $y \neq 0$, which by definition means λ is an eigenvalue of u. □

Theorem 2: Direct Sum of Eigenspaces

If $\lambda_1, \ldots, \lambda_k$ are distinct eigenvalues of u, then their corresponding eigenspaces $E_{\lambda_1}, \ldots, E_{\lambda_k}$ are in direct sum.

PROOF. Suppose for the sake of contradiction they are not in direct sum. Then there exist vectors $x_i \in E_{\lambda_i}$ (not all zero) such that $x_1 + x_2 + \cdots + x_k = 0$. We can assume without loss of generality that all x_i in this sum are non-zero.
If we apply u to this sum, we get $0 = \sum_{i=1}^{k} u(x_i) = \sum_{i=1}^{k} \lambda_i x_i$. Applying u repeatedly l times yields:

$$\sum_{i=1}^{k} u^l(x_i) = \sum_{i=1}^{k} \lambda_i^l x_i = 0$$

We can write the system of equations for $l = 0, \ldots, k-1$ in matrix form:

$$\begin{pmatrix} 1 & 1 & \ldots & 1 \\ \lambda_1 & \lambda_2 & \ldots & \lambda_k \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_1^{k-1} & \lambda_2^{k-1} & \ldots & \lambda_k^{k-1} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

The matrix on the left is the Vandermonde matrix. Its determinant is $\prod_{1 \leq i < j \leq k} (\lambda_j - \lambda_i)$. Since the eigenvalues λ_i are distinct, this determinant is non-zero, meaning the matrix is invertible. Multiplying by the inverse yields $x_1 = x_2 = \cdots = x_k = 0$, which contradicts our assumption that the vectors were non-zero. □

Theorem 3: Vandermonde Determinant

For any scalars $\lambda_1, \ldots, \lambda_k \in K$, the Vandermonde determinant is:

$$V(\lambda_1, \ldots, \lambda_k) = \prod_{1 \leq i < j \leq k} (\lambda_j - \lambda_i)$$

PROOF. We proceed by induction on k. For $k = 2$, the determinant is $\begin{vmatrix} 1 & 1 \\ \lambda_1 & \lambda_2 \end{vmatrix} = \lambda_2 - \lambda_1$, which matches the formula.
Assume the formula holds for $k - 1$. We want to prove it for k. Consider the determinant as a

polynomial $P(X)$ by replacing λ_k with a variable X:

$$P(X) = \begin{vmatrix} 1 & \dots & 1 & 1 \\ \lambda_1 & \dots & \lambda_{k-1} & X \\ \vdots & \ddots & \vdots & \vdots \\ \lambda_1^{k-1} & \dots & \lambda_{k-1}^{k-1} & X^{k-1} \end{vmatrix}$$

Expanding along the last column, we see that $P(X)$ is a polynomial of degree at most $k-1$. If we evaluate $P(X)$ at any $X = \lambda_i$ (for $1 \leq i \leq k-1$), two columns of the matrix become identical, so the determinant is zero. Thus, $\lambda_1, \dots, \lambda_{k-1}$ are the roots of $P(X)$, so we can factor it as:

$$P(X) = c(X - \lambda_1)(X - \lambda_2)\dots(X - \lambda_{k-1})$$

The leading coefficient c corresponds to the X^{k-1} term. Expanding the determinant, this coefficient is exactly the $(k-1) \times (k-1)$ Vandermonde determinant $V(\lambda_1, \dots, \lambda_{k-1})$. By the induction hypothesis, $c = \prod_{1 \leq i < j \leq k-1}(\lambda_j - \lambda_i)$. Evaluating $P(\lambda_k)$ completes the product for k, proving the formula. □

7.1 Diagonalizability

Example 4: Eigenspaces and Direct Sums

1. **Projection Matrix:** Let $M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. The spectrum is $\mathrm{Sp}(M) = \{0, 1\}$. The minimal polynomial is $P_M(X) = X(X-1)$. The eigenspaces are $E_0 = \langle e_3 \rangle$ and $E_1 = \langle e_1, e_2 \rangle$. We see that $E_0 \oplus E_1 = \mathbb{R}^3$.
2. **Distinct Eigenvalues:** Let $M = \begin{pmatrix} \alpha & 0 & 0 \\ 0 & \beta & 0 \\ 0 & 0 & 1 \end{pmatrix}$ with $0 < \beta < \alpha < 1$. The spectrum is $\mathrm{Sp}(M) = \{\alpha, \beta, 1\}$. The minimal polynomial is $P_M(X) = (X-\alpha)(X-\beta)(X-1)$. The eigenspaces are $E_\alpha = \langle e_1 \rangle$, $E_\beta = \langle e_2 \rangle$, and $E_1 = \langle e_3 \rangle$. Their direct sum is $E_\alpha \oplus E_\beta \oplus E_1 = \mathbb{R}^3$.

Theorem 5: Equivalent Conditions for Diagonalizability

For an endomorphism $u \in \mathcal{L}(E)$, the following four conditions are equivalent:

1. $P_u(X)$ is a product of distinct factors of degree 1.
2. The direct sum of the eigenspaces spans the entire space E, i.e., $E = E_{\lambda_1} \oplus \dots \oplus E_{\lambda_k}$.
3. There exists a basis $\mathcal{E}$ such that the matrix $\mathrm{Mat}_{\mathcal{E}}(u)$ is diagonal.
4. u is said to be **diagonalizable** (or "Dz").

Definition 6: Diagonalizable Matrix

A matrix $M \in \mathcal{M}_N(K)$ is diagonalizable if it represents a diagonalizable endomorphism in some basis. Equivalently, it is diagonalizable if there exists an invertible change of basis matrix $P \in GL_N(K)$ such that $P^{-1}MP$ is a diagonal matrix.